

Leonardo Franchi

PhD Student in Mathematics
University of Cambridge
lf511@cam.ac.uk
leo-franchi.github.io

Curriculum Vitae

Education

- Oct. 2025–present **PhD in Mathematics**, University of Cambridge, Cambridge, UK.
Supervisor: Timothy Gowers.
- Oct. 2023–Jul. 2025 **Master's Degree in Mathematics**, University of Padua, Padua, Italy.
Final grade: 110/110 cum laude. Thesis: *Osgood meets DiPerna–Lions*. Advisor: Guido De Philippis.
- Oct. 2020–Oct. 2025 **Galilean Diploma**, Galilean School of Higher Education, University of Padua, Padua, Italy.
- Oct. 2020–Feb. 2023 **Bachelor's Degree in Mathematics**, University of Padua, Padua, Italy.
Final grade: 110/110 cum laude.

Submitted papers and preprints

1. Guido De Philippis and Leonardo Franchi, *Osgood meets Ambrosio–DiPerna–Lions*, arXiv:2607.28118 (2026).
2. Valentino Badalucco and Leonardo Franchi, *Fractal uncertainty principle over $\mathbb{Q}_p$* , arXiv:2607.11534 (2026).
3. Leonardo Franchi, W. T. Gowers and Fredy Yip, *Product-free subsets of $(0, 1)$* , arXiv:2607.06073 (2026).
4. Gaia Carenini and Leonardo Franchi, *A unified abstract regularity lemma*, arXiv:2606.06192 (2026).
5. Gaia Carenini and Leonardo Franchi, *A strengthening of Chang's lemma*, arXiv:2605.07916 (2026).

Invited Talks

- 12 Oct. 2026 *Product-free subsets of $(0, 1)$ and a strengthened Brunn–Minkowski inequality*.
Junior Number Theory Seminar, University of Oxford, Oxford, UK.
- 29 Sep. 2026 *Product-free subsets of $(0, 1)$ and a strengthened Brunn–Minkowski inequality*.
Combinatorics Seminar, University of Bristol, Bristol, UK.
- 26–28 Jun. 2026 *A strengthening of Chang's lemma*.
WOMBL 18 (Warwick–Oxbridge–Manchester–Bristol–Lancaster Meeting on Additive Combinatorics and Analytic Number Theory), Ambleside Campus, University of Cumbria, UK.

Awards and Funding

- 2025-2029 Trinity Cambridge Research Studentship, supported by the Isaac Newton Trust.
- 2024 First Prize, International Mathematics Competition for University Students, Blagoevgrad, Bulgaria.
- 2020 First place, Italian Mathematical Olympiad.

Teaching and Academic Experiences

- Michaelmas 2026 **Undergraduate Supervisor, Optimisation**, Queens' College, University of Cambridge.
- Lent 2026 **Undergraduate Supervisor, Probability (Part IA)**, Queens' College, University of Cambridge.
- Dec. 2025 **Examiner for Undergraduate Mathematics Admissions**, Queens' College, University of Cambridge.
- Jun. 2023, 2024 **Lecturer, International Mathematics Summer Camp**, Beijing Institute of Mathematical Sciences and Applications (BIMSA), Beijing, China.